# Supplementary Information

Coupled Condensation–Melting Heat Transfer and Thermal Design Limits of a Steam-Assisted Pipeline Cleaning Crawler: A Probabilistic Modeling Study

Leon Sandler

Independent Researcher, Northbrook, Illinois, USA. sandler.leon@gmail.com

## S1. Thermal sweep and Monte Carlo behavior (first suite)

| Aerogel (mm) | Atmospheric, 45 min | Atmospheric, 90 min | Evacuated, 45 min | Evacuated, 90 min |
|---|---|---|---|---|
| **5** | 0.1% (0.0–0.7%) | 0% | 51.9% (48.4–55.3%) | 0% |
| **10** | 19.1% (16.5–22.0%) | 0% | 92.4% (90.3–94.0%) | 12% |
| **15** | 62.4% (59.0–65.7%) | 0% | 97.1% (95.7–98.1%) | 35% |
| **20** | 86.5% (84.0–88.7%) | 1% | 98.6% (97.6–99.2%) | 50% |
| **30** | 96.2% (94.7–97.4%) | 16% | 98.9% (97.9–99.4%) | 69% |
| **40** | 99.1% (98.2–99.6%) | 40% | 99.6% (98.9–99.9%) | 80% |
| **50** | 99.2% (98.4–99.7%) | 60% | 99.6% (98.9–99.9%) | 80% |
| **60** | 100.0% (99.5–100.0%) | 72% | 98.9% (97.9–99.4%) | 79% |

*Table S1. P(pack ≤ 55 °C) against aerogel thickness, 45 min (Wilson 95% interval) and 90 min.*

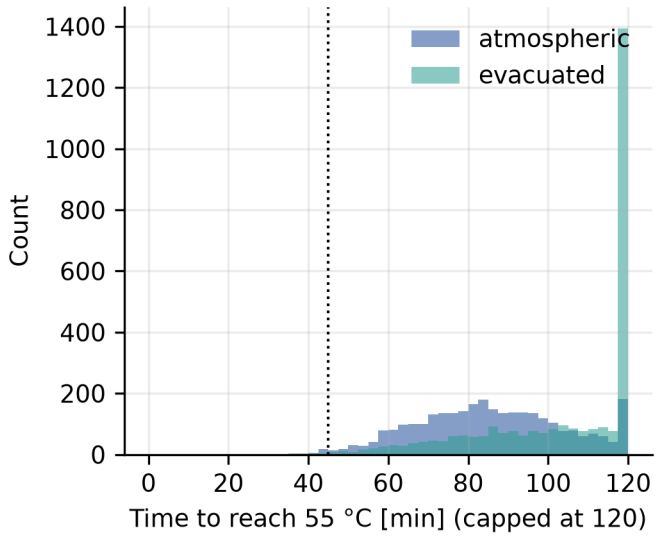



*Fig. S1. Time to reach 55 °C for atmospheric and evacuated aerogel, 40 mm (capped at 120 min).*

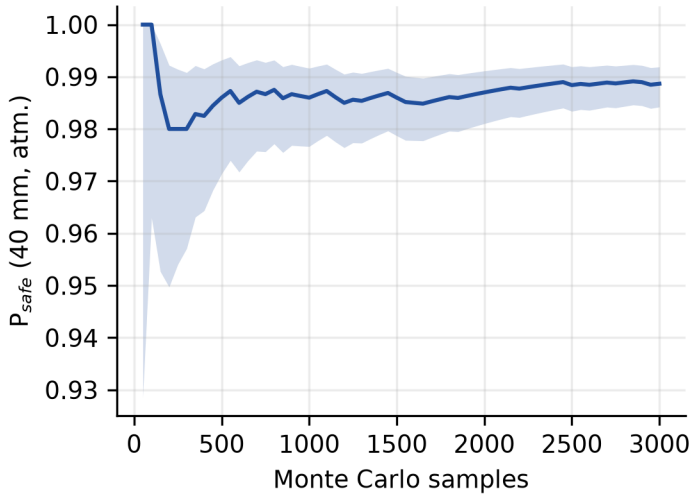



*Fig. S2. Convergence of the Monte Carlo estimate of P(pack ≤ 55 °C) with the number of draws (40 mm, atmospheric); band: Wilson 95% interval.*

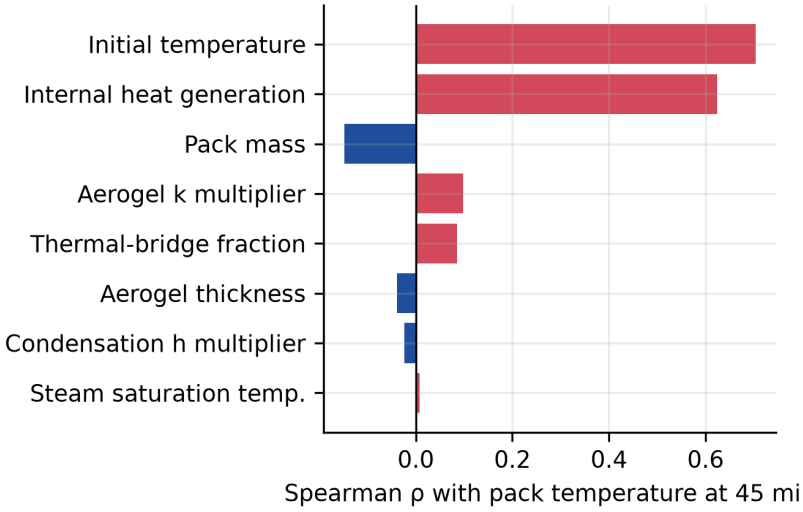



*Fig. S3. Spearman rank correlation of each input with the 45 min pack temperature (first suite; global Sobol indices are in Fig. 9 of the main text).*

## S2. Benchmarks: all cases

| h (W/m²·K) | Thickness (mm) | Ablation formula | Ablation solver | Error (%) | No-drainage formula | No-drainage solver | Error (%) |
|---|---|---|---|---|---|---|---|
| **1,000** | 3 | 4.91 | 4.92 | −0.2 | 54.0 | 65.7 | −17.9 |
| **1,000** | 6 | 9.81 | 9.83 | −0.2 | 206.0 | 251.1 | −17.9 |
| **1,000** | 12 | 19.62 | 19.65 | −0.2 | 804.4 | 980.5 | −18.0 |
| **3,000** | 3 | 1.64 | 1.66 | −1.3 | 50.7 | 61.8 | −18.0 |
| **3,000** | 6 | 3.27 | 3.31 | −1.3 | 199.5 | 243.1 | −18.0 |
| **3,000** | 12 | 6.54 | 6.63 | −1.3 | 791.4 | 964.6 | −18.0 |
| **10,000** | 3 | 0.49 | 0.51 | −4.6 | 49.5 | 60.4 | −18.0 |
| **10,000** | 6 | 0.98 | 1.03 | −4.6 | 197.2 | 240.4 | −18.0 |
| **10,000** | 12 | 1.96 | 2.06 | −4.6 | 786.8 | 959.0 | −18.0 |

*Table S2. Melt time (s) from the formulas and from the enthalpy-method solver (equal properties in both phases, T0 = 25 °C, Tm = 58 °C, steam at 198 °C, no heat-coupling loss). Errors are formula minus solver, in percent of the solver.*

| Cell size (µm) | Melt time (s) |
|---|---|
| **16** | 3.650 |
| **8** | 3.424 |
| **4** | 3.313 |
| **2** | 3.311 |

*Table S3. Ablation melt time of the enthalpy solver against cell size (h = 3 kW/m²·K, 6 mm).*

## S3. Global sensitivity: Sobol indices with 95% bootstrap intervals

| Input | First order | Total |
|---|---|---|
| **aerogel k multiplier** | 0.02 (−0.07 to 0.14) | 0.02 (0.02 to 0.02) |
| **aerogel thickness** | 0.00 (−0.04 to 0.04) | 0.00 (0.00 to 0.00) |
| **thermal-bridge fraction** | 0.01 (−0.06 to 0.07) | 0.01 (0.01 to 0.01) |
| **internal heat generation** | 0.46 (−0.18 to 1.10) | 0.48 (0.42 to 0.55) |
| **pack mass** | 0.02 (−0.13 to 0.16) | 0.03 (0.02 to 0.03) |
| **initial temperature** | 0.49 (−0.02 to 1.20) | 0.49 (0.43 to 0.55) |
| **steam temperature** | 0.00 (−0.01 to 0.01) | 0.00 (0.00 to 0.00) |
| **condensation multiplier** | 0.00 (0.00 to 0.00) | 0.00 (0.00 to 0.00) |

*Table S4. Sobol indices of the pack temperature at 45 min (N = 512 base samples).*

| Input | First order | Total |
|---|---|---|
| **deposit thickness** | 0.24 (0.19 to 0.28) | 0.24 (0.23 to 0.24) |
| **melt point** | 0.01 (0.00 to 0.02) | 0.01 (0.01 to 0.01) |
| **deposit density** | 0.00 (0.00 to 0.01) | 0.00 (0.00 to 0.00) |
| **latent heat** | 0.01 (0.00 to 0.02) | 0.01 (0.01 to 0.01) |
| **pipe temperature** | 0.00 (0.00 to 0.01) | 0.00 (0.00 to 0.00) |
| **heat-coupling fraction** | 0.11 (0.07 to 0.14) | 0.11 (0.10 to 0.11) |
| **non-condensable factor** | 0.00 (0.00 to 0.01) | 0.00 (0.00 to 0.00) |
| **contact length** | 0.07 (0.05 to 0.10) | 0.07 (0.07 to 0.07) |
| **steam temperature** | 0.00 (0.00 to 0.00) | 0.00 (0.00 to 0.00) |
| **condensation multiplier** | 0.00 (0.00 to 0.01) | 0.00 (0.00 to 0.00) |
| **retained film** | 0.55 (0.49 to 0.62) | 0.55 (0.54 to 0.57) |

*Table S4. Sobol indices of the log10 range, retained film lognormal (N = 8,192 base samples).*

| Input | First order | Total |
|---|---|---|
| **deposit thickness** | 0.45 (0.30 to 0.61) | 0.48 (0.47 to 0.50) |
| **melt point** | 0.02 (−0.01 to 0.06) | 0.02 (0.02 to 0.02) |
| **deposit density** | 0.01 (−0.01 to 0.02) | 0.00 (0.00 to 0.00) |
| **latent heat** | 0.02 (−0.02 to 0.06) | 0.03 (0.02 to 0.03) |
| **pipe temperature** | 0.00 (−0.01 to 0.02) | 0.00 (0.00 to 0.00) |
| **heat-coupling fraction** | 0.22 (0.11 to 0.32) | 0.23 (0.23 to 0.24) |
| **non-condensable factor** | 0.06 (−0.01 to 0.11) | 0.08 (0.08 to 0.08) |
| **contact length** | 0.14 (0.06 to 0.22) | 0.16 (0.15 to 0.16) |
| **steam temperature** | 0.00 (0.00 to 0.01) | 0.00 (0.00 to 0.00) |
| **condensation multiplier** | 0.04 (−0.01 to 0.09) | 0.05 (0.04 to 0.05) |

*Table S4. Sobol indices of the log10 range, instant removal (N = 8,192 base samples).*

| Input | First order | Total |
|---|---|---|
| **deposit thickness** | 0.65 (0.61 to 0.68) | 0.65 (0.63 to 0.67) |
| **melt point** | 0.01 (0.01 to 0.02) | 0.01 (0.01 to 0.01) |

| Input | First order | Total |
|---|---|---|
| **deposit density** | 0.01 (0.00 to 0.01) | 0.01 (0.01 to 0.01) |
| **latent heat** | 0.03 (0.03 to 0.04) | 0.03 (0.03 to 0.04) |
| **pipe temperature** | 0.00 (0.00 to 0.01) | 0.00 (0.00 to 0.00) |
| **heat-coupling fraction** | 0.29 (0.27 to 0.31) | 0.29 (0.28 to 0.30) |
| **non-condensable factor** | 0.00 (0.00 to 0.00) | 0.00 (0.00 to 0.00) |
| **contact length** | 0.00 (0.00 to 0.00) | 0.00 (0.00 to 0.00) |
| **steam temperature** | 0.00 (0.00 to 0.00) | 0.00 (0.00 to 0.00) |
| **condensation multiplier** | 0.00 (0.00 to 0.00) | 0.00 (0.00 to 0.00) |
| **retained film** | 0.00 (0.00 to 0.00) | 0.00 (0.00 to 0.00) |

*Table S4. Sobol indices of the log10 melting steam per metre, retained film lognormal (N = 8,192 base samples).*

## S4. Correlated inputs

| ρ (T0, P_gen) | ρ (thickness, melt point) | P(pack ≤ 55 °C) (95% interval) | Median range (m) | P(range ≥ 100 m) |
|---|---|---|---|---|
| **0.0** | 0.0 | 99.1% (98.7–99.4%) | 133.1 | 64% |
| **0.6** | 0.0 | 97.4% (96.7–97.9%) | 132.9 | 64% |
| **−0.6** | 0.0 | 99.8% (99.6–99.9%) | 133.4 | 64% |
| **0.0** | 0.6 | 99.1% (98.7–99.4%) | 133.8 | 63% |
| **0.0** | −0.6 | 99.1% (98.7–99.4%) | 133.6 | 65% |
| **0.6** | 0.6 | 97.4% (96.7–97.9%) | 133.5 | 63% |

*Table S5. Gaussian-copula correlation of starting temperature with heat generation and of deposit thickness with melt point (3,000 draws).*

## S5. Dynamic film and design search

| Case | Median range (m) | Median film (mm) | P(range ≥ 100 m) |
|---|---|---|---|
| **baseline** | 17.7 | 0.35 | 0.3% |
| **viscosity x0.25** | 23.9 | 0.24 | 1.4% |
| **viscosity x4** | 13.1 | 0.50 | 0.1% |
| **drain path x0.25** | 23.9 | 0.24 | 1.4% |
| **drain path x4** | 13.1 | 0.50 | 0.1% |
| **residual film 0.02 mm** | 17.1 | 0.36 | 0.2% |
| **residual film 0.05 mm** | 16.2 | 0.39 | 0.2% |
| **steam shear 0.1 Pa** | 18.0 | 0.34 | 0.3% |
| **steam shear 1 Pa** | 20.5 | 0.29 | 0.7% |
| **steam shear 10 Pa** | 32.8 | 0.16 | 4.2% |
| **wall near the invert (sin 0.2)** | 13.5 | 0.48 | 0.1% |
| **wall at the spring line (sin 1.0)** | 19.2 | 0.32 | 0.5% |

*Table S6. Dynamic film: sensitivity of the median range and of the final film to the drainage assumptions.*

| Jacket | Aerogel (mm) | Mission (min) | Set-point (m/s) | Hull OD (mm) | P(≤ 55 °C) | Range (m) | Steam (kg/m) | Energy (Wh) |
|---|---|---|---|---|---|---|---|---|
| **evacuated** | 40 | 60 | 0.20 | 216 | 97.0% | 177.7 | 1.73 | 69 |
| **evacuated** | 35 | 60 | 0.20 | 206 | 96.5% | 177.6 | 1.73 | 69 |
| **evacuated** | 30 | 60 | 0.20 | 196 | 95.7% | 177.3 | 1.73 | 69 |
| **atmospheric** | 40 | 45 | 0.20 | 216 | 99.1% | 133.7 | 1.73 | 53 |
| **atmospheric** | 35 | 45 | 0.20 | 206 | 98.7% | 133.6 | 1.73 | 53 |
| **evacuated** | 25 | 45 | 0.20 | 186 | 99.1% | 133.6 | 1.73 | 52 |
| **evacuated** | 30 | 45 | 0.20 | 196 | 99.3% | 133.6 | 1.73 | 52 |
| **evacuated** | 35 | 45 | 0.20 | 206 | 99.5% | 133.6 | 1.73 | 52 |
| **evacuated** | 40 | 45 | 0.20 | 216 | 99.6% | 133.6 | 1.73 | 52 |
| **evacuated** | 20 | 45 | 0.20 | 176 | 98.7% | 133.5 | 1.73 | 52 |
| **evacuated** | 15 | 45 | 0.20 | 166 | 96.7% | 133.1 | 1.73 | 52 |

| Jacket | Aerogel (mm) | Mission (min) | Set-point (m/s) | Hull OD (mm) | P(≤ 55 °C) | Range (m) | Steam (kg/m) | Energy (Wh) |
|---|---|---|---|---|---|---|---|---|
| **evacuated** | 25 | 30 | 0.20 | 186 | 100.0% | 89.2 | 1.73 | 35 |
| **atmospheric** | 20 | 30 | 0.20 | 176 | 99.8% | 89.1 | 1.73 | 35 |
| **evacuated** | 15 | 30 | 0.20 | 166 | 99.8% | 89.1 | 1.73 | 35 |
| **evacuated** | 20 | 30 | 0.20 | 176 | 100.0% | 89.1 | 1.73 | 35 |
| **evacuated** | 10 | 30 | 0.20 | 156 | 99.2% | 89.1 | 1.73 | 35 |

*Table S7. Pareto-optimal feasible designs for instant removal (up to 20, by range).*

| Jacket | Aerogel (mm) | Mission (min) | Set-point (m/s) | Hull OD (mm) | P(≤ 55 °C) | Range (m) | Steam (kg/m) | Energy (Wh) |
|---|---|---|---|---|---|---|---|---|
| **evacuated** | 40 | 60 | 0.05 | 216 | 97.0% | 23.6 | 1.73 | 69 |
| **evacuated** | 40 | 60 | 0.10 | 216 | 97.0% | 23.6 | 1.73 | 69 |
| **evacuated** | 40 | 60 | 0.20 | 216 | 97.0% | 23.6 | 1.73 | 69 |
| **evacuated** | 35 | 60 | 0.05 | 206 | 96.5% | 23.5 | 1.73 | 69 |
| **evacuated** | 35 | 60 | 0.10 | 206 | 96.5% | 23.5 | 1.73 | 69 |
| **evacuated** | 35 | 60 | 0.20 | 206 | 96.5% | 23.5 | 1.73 | 69 |
| **evacuated** | 30 | 60 | 0.05 | 196 | 95.7% | 23.4 | 1.73 | 69 |
| **evacuated** | 30 | 60 | 0.10 | 196 | 95.7% | 23.4 | 1.73 | 69 |
| **evacuated** | 30 | 60 | 0.20 | 196 | 95.7% | 23.4 | 1.73 | 69 |
| **evacuated** | 35 | 45 | 0.05 | 206 | 99.5% | 17.8 | 1.73 | 52 |
| **evacuated** | 35 | 45 | 0.10 | 206 | 99.5% | 17.8 | 1.73 | 52 |
| **evacuated** | 35 | 45 | 0.20 | 206 | 99.5% | 17.8 | 1.73 | 52 |
| **evacuated** | 40 | 45 | 0.05 | 216 | 99.6% | 17.8 | 1.73 | 52 |
| **evacuated** | 40 | 45 | 0.10 | 216 | 99.6% | 17.8 | 1.73 | 52 |
| **evacuated** | 40 | 45 | 0.20 | 216 | 99.6% | 17.8 | 1.73 | 52 |
| **evacuated** | 25 | 45 | 0.05 | 186 | 99.1% | 17.7 | 1.73 | 52 |
| **evacuated** | 25 | 45 | 0.10 | 186 | 99.1% | 17.7 | 1.73 | 52 |
| **evacuated** | 25 | 45 | 0.20 | 186 | 99.1% | 17.7 | 1.73 | 52 |
| **evacuated** | 30 | 45 | 0.05 | 196 | 99.3% | 17.7 | 1.73 | 52 |
| **evacuated** | 30 | 45 | 0.10 | 196 | 99.3% | 17.7 | 1.73 | 52 |

*Table S7. Pareto-optimal feasible designs for dynamic film (up to 20, by range).*

| NPS (in) | 0 | 10 | 20 | 30 | 40 | 50 |
|---|---|---|---|---|---|---|
| **8** | 18% | 4% | 1% | 0% | 0% | 0% |
| **10** | 96% | 87% | 62% | 19% | 2% | 0% |
| **12** | 100% | 100% | 99% | 97% | 58% | 4% |
| **14** | 100% | 100% | 99% | 97% | 58% | 4% |
| **16** | 100% | 100% | 99% | 97% | 58% | 4% |

*Table S8. Feasible envelope: P(pack ≤ 55 °C, 45 min) by nominal pipe size (rows) and pipe temperature (columns, °C), 400 draws per cell.*

## S6. Pipe length and logistics (all cases)

| Mission | Melt removal | L (m) | P(one) | Dep. median | Dep. p95 | Active (h) | Total (h) | Range (m) |
|---|---|---|---|---|---|---|---|---|
| **45 min atmospheric** | ablation | 0.3 | 100% | 1 | 1 | 0.00 | 11.5 (8.3–14.7) | 133.6 |
| **45 min atmospheric** | ablation | 10 | 100% | 1 | 1 | 0.06 | 11.6 (8.4–14.8) | 133.6 |
| **45 min atmospheric** | ablation | 100 | 65% | 1 | 3 | 0.56 | 12.9 (9.1–18.4) | 133.6 |
| **45 min atmospheric** | ablation | 300 | 0% | 3 | 8 | 1.68 | 17.1 (12.0–31.5) | 133.6 |
| **45 min atmospheric** | ablation | 600 | 0% | 5 | 16 | 3.36 | 22.6 (14.9–51.9) | 133.6 |
| **45 min atmospheric** | ablation | 1,000 | 0% | 8 | 26 | 5.61 | 30.6 (18.2–79.6) | 133.6 |
| **45 min atmospheric** | retained 0.5 mm | 0.3 | 100% | 1 | 1 | 0.02 | 11.5 (8.3–14.7) | 12.5 |
| **45 min atmospheric** | retained 0.5 mm | 10 | 62% | 1 | 3 | 0.60 | 13.1 (9.1–18.5) | 12.5 |
| **45 min atmospheric** | retained 0.5 mm | 100 | 0% | 9 | 27 | 6.00 | 32.1 (16.8–81.2) | 12.5 |
| **45 min atmospheric** | retained 0.5 mm | 300 | 0% | 25 | 79 | 18.01 | 75.3 (30.6–224.9) | 12.5 |

| Mission | Melt removal | L (m) | P(one) | Dep. median | Dep. p95 | Active (h) | Total (h) | Range (m) |
|---|---|---|---|---|---|---|---|---|
| **45 min atmospheric** | retained 0.5 mm | 600 | 0% | 49 | 158 | 36.02 | 140.0 (50.0–438.8) | 12.5 |
| **45 min atmospheric** | retained 0.5 mm | 1,000 | 0% | 81 | 263 | 60.03 | 227.1 (76.3–723.3) | 12.5 |
| **45 min atmospheric** | no drainage | 0.3 | 97% | 1 | 1 | 0.10 | 11.7 (8.5–15.1) | 2.3 |
| **45 min atmospheric** | no drainage | 10 | 9% | 5 | 28 | 3.30 | 22.7 (11.6–82.2) | 2.3 |
| **45 min atmospheric** | no drainage | 100 | 0% | 45 | 274 | 32.99 | 130.5 (29.2–725.2) | 2.3 |
| **45 min atmospheric** | no drainage | 300 | 0% | 133 | 822 | 98.96 | 369.0 (66.2–2,154.9) | 2.3 |
| **45 min atmospheric** | no drainage | 600 | 0% | 265 | 1,643 | 197.92 | 727.6 (121.5–4,301.2) | 2.3 |
| **45 min atmospheric** | no drainage | 1,000 | 0% | 442 | 2,738 | 329.86 | 1,206.4 (195.9–7,162.3) | 2.3 |
| **90 min evacuated** | ablation | 0.3 | 100% | 1 | 1 | 0.00 | 11.5 (8.3–14.7) | 255.2 |
| **90 min evacuated** | ablation | 10 | 100% | 1 | 1 | 0.06 | 11.6 (8.4–14.8) | 255.2 |
| **90 min evacuated** | ablation | 100 | 89% | 1 | 2 | 0.56 | 12.4 (9.0–16.2) | 255.2 |
| **90 min evacuated** | ablation | 300 | 41% | 2 | 5 | 1.68 | 14.9 (10.1–24.5) | 255.2 |
| **90 min evacuated** | ablation | 600 | 0% | 3 | 9 | 3.37 | 18.8 (13.0–38.2) | 255.2 |
| **90 min evacuated** | ablation | 1,000 | 0% | 4 | 14 | 5.61 | 23.9 (15.0–57.0) | 255.2 |
| **90 min evacuated** | retained 0.5 mm | 0.3 | 100% | 1 | 1 | 0.02 | 11.5 (8.3–14.7) | 23.8 |
| **90 min evacuated** | retained 0.5 mm | 10 | 89% | 1 | 2 | 0.60 | 12.4 (9.0–16.3) | 23.8 |
| **90 min evacuated** | retained 0.5 mm | 100 | 2% | 5 | 14 | 6.02 | 24.9 (14.4–57.8) | 23.8 |
| **90 min evacuated** | retained 0.5 mm | 300 | 0% | 13 | 42 | 18.05 | 53.4 (23.9–151.6) | 23.8 |
| **90 min evacuated** | retained 0.5 mm | 600 | 0% | 26 | 84 | 36.10 | 95.9 (37.1–292.9) | 23.8 |
| **90 min evacuated** | retained 0.5 mm | 1,000 | 0% | 43 | 140 | 60.17 | 152.5 (54.9–480.9) | 23.8 |
| **90 min evacuated** | no drainage | 0.3 | 99% | 1 | 1 | 0.10 | 11.7 (8.4–14.9) | 4.3 |
| **90 min evacuated** | no drainage | 10 | 23% | 3 | 15 | 3.29 | 18.6 (10.6–58.7) | 4.3 |
| **90 min evacuated** | no drainage | 100 | 0% | 24 | 146 | 32.89 | 88.6 (23.2–498.1) | 4.3 |
| **90 min evacuated** | no drainage | 300 | 0% | 70 | 438 | 98.66 | 245.1 (47.9–1,473.8) | 4.3 |
| **90 min evacuated** | no drainage | 600 | 0% | 138 | 875 | 197.31 | 479.7 (84.6–2,936.8) | 4.3 |
| **90 min evacuated** | no drainage | 1,000 | 0% | 230 | 1,458 | 328.86 | 792.3 (134.8–4,891.9) | 4.3 |
| **45 min atmospheric** | dynamic film | 0.3 | 100% | 1 | 1 | 0.01 | 11.5 (8.3–14.7) | 17.7 |
| **45 min atmospheric** | dynamic film | 10 | 80% | 1 | 2 | 0.42 | 12.4 (8.9–16.6) | 17.7 |
| **45 min atmospheric** | dynamic film | 100 | 0% | 6 | 18 | 4.22 | 25.9 (14.8–57.3) | 17.7 |
| **45 min atmospheric** | dynamic film | 300 | 0% | 17 | 52 | 12.66 | 56.7 (25.0–150.5) | 17.7 |
| **45 min atmospheric** | dynamic film | 600 | 0% | 34 | 104 | 25.33 | 102.8 (39.1–291.6) | 17.7 |
| **45 min atmospheric** | dynamic film | 1,000 | 0% | 57 | 172 | 42.21 | 164.4 (58.2–479.6) | 17.7 |
| **90 min evacuated** | dynamic film | 0.3 | 100% | 1 | 1 | 0.01 | 11.5 (8.3–14.7) | 33.7 |
| **90 min evacuated** | dynamic film | 10 | 96% | 1 | 1 | 0.42 | 12.1 (8.8–15.5) | 33.7 |
| **90 min evacuated** | dynamic film | 100 | 5% | 3 | 10 | 4.22 | 20.8 (12.8–41.7) | 33.7 |
| **90 min evacuated** | dynamic film | 300 | 0% | 9 | 28 | 12.66 | 41.0 (19.9–104.1) | 33.7 |
| **90 min evacuated** | dynamic film | 600 | 0% | 18 | 56 | 25.33 | 71.2 (29.8–198.0) | 33.7 |
| **90 min evacuated** | dynamic film | 1,000 | 0% | 30 | 93 | 42.21 | 111.7 (42.6–321.9) | 33.7 |

*Table S9. Deployments, active time and total intervention time by mission, melt-removal case and pipe length (median; total with 5th–95th percentile).*

## S7. Pipe size and temperature (detail)

| NPS (in) | Bore (mm) | Hull OD (mm) | Pack radius (mm) | Aerogel (mm) | P atm. | P evac. | h scale | Steam scale |
|---|---|---|---|---|---|---|---|---|
| **4** | 102.3 | – | 25 | does not fit | – | – | 1.31 | 0.34 |
| **6** | 154.1 | 65 | 25 | does not fit | – | – | 1.19 | 0.51 |
| **8** | 202.7 | 114 | 25 | 24 | 0.0% (0.0–0.5%) | 5.1% (3.8–6.9%) | 1.11 | 0.67 |
| **10** | 254.5 | 166 | 35 | 40 | 42.2% (38.9–45.7%) | 50.0% (46.5–53.5%) | 1.05 | 0.83 |
| **12** | 304.8 | 216 | 60 | 40 | 99.2% (98.4–99.7%) | 99.4% (98.5–99.7%) | 1.00 | 1.00 |
| **14** | 336.5 | 248 | 60 | 40 | 99.2% (98.4–99.7%) | 99.4% (98.5–99.7%) | 0.98 | 1.10 |
| **16** | 387.3 | 298 | 60 | 40 | 99.2% (98.4–99.7%) | 99.4% (98.5–99.7%) | 0.94 | 1.27 |

*Table S10. Fit and thermal margin by nominal pipe size (800 draws; Wilson interval). h scale: condensation coefficient relative to 12 in; steam scale: melting steam per metre relative to 12 in.*

| Temperature (°C) | P atm. (95% interval) | P evac. (95% interval) | Median pack T at 45 min (°C) | Melt time (s) | Range (m) |
|---|---|---|---|---|---|
| **0** | 100.0% (99.5–100.0%) | 100.0% (99.5–100.0%) | 14.3 | 13.1 | 111 |
| **10** | 100.0% (99.5–100.0%) | 100.0% (99.5–100.0%) | 24.2 | 12.3 | 119 |
| **25** | 98.8% (97.7–99.3%) | 99.9% (99.3–100.0%) | 39.1 | 10.9 | 133 |
| **35** | 84.5% (81.8–86.8%) | 89.8% (87.5–91.7%) | 49.0 | 10.1 | 139 |
| **45** | 21.9% (19.1–24.9%) | 35.9% (32.6–39.3%) | 58.9 | 9.3 | 102 |
| **55** | 0.4% (0.1–1.1%) | 0.1% (0.0–0.7%) | 68.7 | 8.4 | 0 |

*Table S11. Pipe / fluid temperature (12 in design, 40 mm aerogel; 800 draws): probability, median pack temperature at 45 min (atmospheric), median melt time and range.*

## S8. Bypass: peak speed by pocket volume

| $V_g$ (m³) | 0 | 5 | 10 | 20 | 40 | 80 | 160 | 320 |
|---|---|---|---|---|---|---|---|---|
| **0.3** | 97% | 97% | 96% | 95% | 92% | 88% | 82% | 72% |
| **1.0** | 98% | 98% | 98% | 98% | 97% | 95% | 92% | 87% |
| **3.0** | 99% | 99% | 99% | 98% | 98% | 98% | 96% | 94% |
| **10.0** | 99% | 99% | 99% | 99% | 99% | 99% | 98% | 98% |

*Table S12. P(peak speed > 1 m/s) against bypass area (cm²; columns) for four gas-pocket volumes (rows, m³); other inputs sampled.*

| $V_g$ (m³) | 20 cm² | 80 cm² | 320 cm² |
|---|---|---|---|
| **0.3** | 76 | 19 | 5 |
| **1.0** | 254 | 64 | 16 |
| **3.0** | 762 | 190 | 48 |
| **10.0** | 2,538 | 635 | 159 |

*Table S13. Median time for a fully open bypass to halve the break-through differential (ms), tool held at rest, against pocket volume and bypass area.*

| $Q_s$ (m³/s at upstream conditions) | Bypass area (cm²) |
|---|---|
| **0.01** | 1.6 |
| **0.03** | 4.7 |
| **0.10** | 15.6 |
| **0.30** | 46.9 |
| **1.00** | 156.3 |

*Table S14. Bypass flow area that holds a 0.5 bar differential at 18 bar upstream (assumed methane-like gas, $C_d$ = 0.7) for a supply flow $Q_s$.*

## S9. Maintenance lattice (first suite)

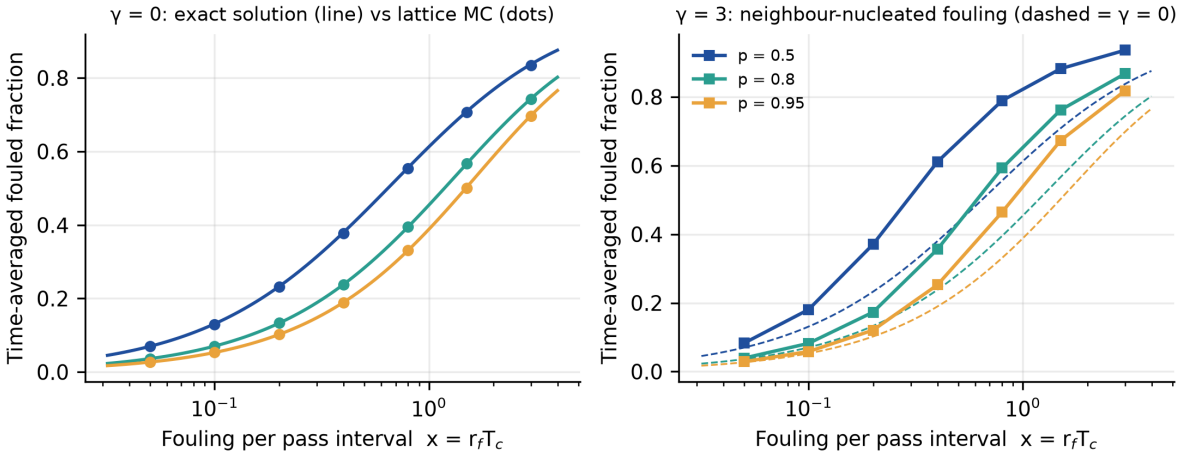



*Fig. S4. Time-averaged fouled fraction against fouling per pass interval for three clearing probabilities: non-interacting case (left) and neighbor-nucleated fouling, γ = 3 (right).*

For $\gamma$=0 the lattice Monte Carlo agrees with the exact solution within 0.26 percentage points. Neighbor-nucleated fouling raises the fouled fraction substantially: for $x$=0.8 and $p$=0.8 it rises from 0.40 to 0.59, so a cleaning interval that suffices for independent fouling may fail when deposits nucleate on existing deposits. The lattice is not calibrated to any deposit growth rate and is reported as a planning illustration only.

## S10. Automated checks

tests/test_all.py runs 62 checks. First suite: unchanged (sha256) and reproduces its published results; the maintenance-lattice formula. Extension: the extended thermal, melt and surge functions equal the first suite when the added mechanisms are off; monotonic range with retained film; the no-drainage limit; the series-resistance flux; the $D^{-1/4}$ scaling of the condensation coefficient; zero and choked orifice flow; gas-fraction bounds; monotonic bypass benefit; venting-time scaling; proportionality of the holding area to the supply flow; the 12 in hull; the fit in 4, 6 and 8 in pipes; the wall-heating energy balance; mesh convergence; the 270 m and 540 m caps. Revision: the enthalpy solver against the exact Neumann solution; the ablation formula and the no-drainage bias against the solver; convergence of the ablation solution; the conduction solver against an independent stiff-ODE integration; the Chato to Nusselt ratio; restoration of the swapped condensation coefficient; the dynamic film in its two limits; the series-resistance identity and the Π_f threshold; IF97 saturation pressures; the Sobol estimator on a function with known indices and the sums of the computed first-order indices; the Pareto mask; the constraints of the design search; the Class 150 limit; the speed-cap accounting; and the sign of the correlation effect.

## References


[1] A. Aiyejina, D.P. Chakrabarti, A. Pilgrim, M. Sastry, Wax formation in oil pipelines: A critical review, International Journal of Multiphase Flow 37 (2011) 671–694. https://doi.org/10.1016/j.ijmultiphaseflow.2011.02.007

[2] Z. Huang, S. Zheng, H.S. Fogler, Wax Deposition: Experimental Characterizations, Theoretical Modeling, and Field Practices, CRC Press, Boca Raton (2016). https://doi.org/10.1201/b18482

[3] S. Kim, J. Jeong, K. Yoo, H. Yoo, Y. Seo, Experimental investigation on speed excursion of PIG due to friction variation in natural gas pipeline, Journal of Natural Gas Science and Engineering 104 (2022) 104659. https://doi.org/10.1016/j.jngse.2022.104659

[4] T.T. Nguyen, S.B. Kim, H.R. Yoo, Y.W. Rho, Modeling and simulation for PIG with bypass flow control in natural gas pipeline, KSME International Journal 15 (2001) 1302–1310. https://doi.org/10.1007/bf03185671

[5] X. Zhu, S. Zhang, G. Tan, D. Wang, W. Wang, Experimental study on dynamics of rotatable bypass-valve in speed control pig in gas pipeline, Measurement 47 (2014) 686–692. https://doi.org/10.1016/j.measurement.2013.08.060

[6] J. Quarini, Ice-pigging to reduce and remove fouling and to achieve clean-in-place, Applied Thermal Engineering 22 (2002) 747–753. https://doi.org/10.1016/s1359-4311(02)00019-4

[7] G. Quarini, E. Ainslie, M. Herbert, T. Deans, D. Ash, D. Rhys, N. Haskins, G. Norton, S. Andrews, M. Smith, Investigation and development of an innovative pigging technique for the water-supply industry, Proceedings of the Institution of Mechanical Engineers, Part E: Journal of Process Mechanical Engineering 224 (2010) 79–89. https://doi.org/10.1243/09544089jpme312

[8] Wazoku (InnoCentive), Innovative Pipeline Cleaning Solutions (scouting challenge for YPF), https://www.innocentive.com/challenges/innovative-pipeline-cleaning-solutions/ (accessed 7 November 2026).

[9] M. Koebel, A. Rigacci, P. Achard, Aerogel-based thermal superinsulation: an overview, Journal of Sol-Gel Science and Technology 63 (2012) 315–339. https://doi.org/10.1007/s10971-012-2792-9

[10] IEC 60079-0, Explosive atmospheres — Part 0: Equipment — General requirements, International Electrotechnical Commission, Geneva (current edition).

[11] X. Feng, M. Ouyang, X. Liu, L. Lu, Y. Xia, X. He, Thermal runaway mechanism of lithium ion battery for electric vehicles: A review, Energy Storage Materials 10 (2018) 246–267. https://doi.org/10.1016/j.ensm.2017.05.013

[12] W. Nusselt, Die Oberflächenkondensation des Wasserdampfes, Zeitschrift des Vereines Deutscher Ingenieure 60 (1916) 541–546, 569–575.

[13] F.P. Incropera, D.P. DeWitt, T.L. Bergman, A.S. Lavine, Fundamentals of Heat and Mass Transfer, 6th ed., Wiley, Hoboken (2007).

[14] J.C. Chato, Laminar condensation inside horizontal and inclined tubes, ASHRAE Journal 4 (1962) 52–60.

[15] M.K. Dobson, J.C. Chato, Condensation in smooth horizontal tubes, Journal of Heat Transfer 120 (1998) 193–213. https://doi.org/10.1115/1.2830043

[16] W. Wagner, A. Kruse, IAPWS Industrial Formulation 1997 for the Thermodynamic Properties of Water and Steam, in: Properties of Water and Steam, Springer, Berlin (1998) 7–37. https://doi.org/10.1007/978-3-662-03529-0_3

[17] L. Sandler, MyUncle: maintenance-lattice simulation framework (software repository), https://github.com/sandlerleon/MyUncle (accessed 7 November 2026).

[18] I.M. Sobol', Global sensitivity indices for nonlinear mathematical models and their Monte Carlo estimates, Mathematics and Computers in Simulation 55 (2001) 271–280. https://doi.org/10.1016/s0378-4754(00)00270-6

[19] A. Saltelli, M. Ratto, T. Andres, F. Campolongo, J. Cariboni, D. Gatelli, M. Saisana, S. Tarantola, Global Sensitivity Analysis: The Primer, Wiley, Chichester (2008). https://doi.org/10.1002/9780470725184

[20] M.J.W. Jansen, Analysis of variance designs for model output, Computer Physics Communications 117 (1999) 35–43. https://doi.org/10.1016/s0010-4655(98)00154-4

[21] A. Saltelli, P. Annoni, I. Azzini, F. Campolongo, M. Ratto, S. Tarantola, Variance based sensitivity analysis of model output. Design and estimator for the total sensitivity index, Computer Physics Communications 181 (2010) 259–270. https://doi.org/10.1016/j.cpc.2009.09.018

[22] N. Metropolis, S. Ulam, The Monte Carlo method, Journal of the American Statistical Association 44 (1949) 335–341. https://doi.org/10.1080/01621459.1949.10483310

[23] E.B. Wilson, Probable inference, the law of succession, and statistical inference, Journal of the American Statistical Association 22 (1927) 209–212. https://doi.org/10.1080/01621459.1927.10502953

[24] V. Alexiades, A.D. Solomon, Mathematical Modeling of Melting and Freezing Processes, CRC Press, Boca Raton (2018; first published by Hemisphere, 1993). https://doi.org/10.1201/9780203749449

[25] A. Yadav, B. Donepudi, S. Siddani, Numerical and experimental investigation of melting characteristics of phase change material-RT58, Thermal Science and Engineering Progress 17 (2020) 100378. https://doi.org/10.1016/j.tsep.2019.100378

[26] A. Sharma, V.V. Tyagi, C.R. Chen, D. Buddhi, Review on thermal energy storage with phase change materials and applications, Renewable and Sustainable Energy Reviews 13 (2009) 318–345. https://doi.org/10.1016/j.rser.2007.10.005

[27] ASME B16.5, Pipe Flanges and Flanged Fittings: NPS 1/2 through NPS 24 Metric/Inch Standard, American Society of Mechanical Engineers, New York (current edition).